KEVIN WU

SOFTWARE DEVELOPER

Contact Info

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ABOUT ME

A software developer who enjoys turning logical ideas into working systems. Through hands-on client project work, I've learned to quickly pick up new tools and adapt to fast-paced environments, and I use Claude Code as an agentic development partner – orchestrating multiple AI agents, automating recurring engineering tasks, and running AI-assisted code and security reviews – to ship full-stack features faster without sacrificing quality.

WORK EXPERIENCE

Multi-tenant Course Scheduling Platform - Personal Project (2026)

Tech Stack: React, TypeScript, Express, PostgreSQL, i18n (3 languages), JWT, Google OAuth (SSO), Stripe, Leaflet / OpenStreetMap

Responsibilities & Achievements:

- Built a multi-company course platform: QR-code onboarding, month-calendar enrolment, cross-company time-conflict checks, role-based staff management, Google SSO sign-in, embedded location maps (Leaflet / OpenStreetMap), and a Stripe billing skeleton.
- Designed a flexible scheduling engine: weekly runs auto-generate an editable per-session table (individual date/time overrides, add or skip sessions) with a one-click "renew term" flow that carries settings forward.
- Developed it end-to-end with an **AI-assisted, multi-agent workflow using Claude Code** – model routing by task complexity, plan-mode design before implementation, and an autonomous weekly task pipeline that self-verifies (typecheck / build) and commits.
- Hardened the backend – per-IP rate limiting, security headers (helmet), JWT fail-fast, user-enumeration protection – and applied **AI-driven code & security review** to catch and fix an account-impersonation vulnerability in the company-registration flow.

Continuous Technologies International Limited - Programmer (10/25 - Now)

SMC Project

Project Description: A customer engagement and routing system integrated with Genesys Cloud to assign communication channels and live agents for chat interactions, auto-replies, and report generation.

Tech Stack: Vue.js, Node.js, Microsoft SQL Server, WebSockets.

Responsibilities & Achievements:

- Developed and maintained the Vue.js frontend interface for live agent chat interactions.
- Implemented WebSockets on both frontend and backend for real-time chat messaging and system notifications.
- Built features for agents to search, view, and log customer profiles and historical data during live interactions.
- Configured routing queues with custom trigger conditions and automated actions for chat distribution.
- Developed Node.js backend logic to batch-export CSV reports and chat histories, generating download paths based on user authorization.
- Created background configuration modules for admins to manage multiple dashboard layouts and UI themes.
- Wrote SQL queries and stored procedures in Microsoft SQL Server to handle system data storage and report calculations.

Server Integration Backend

Tech Stack: Golang, MSSQL

Responsibilities & Achievements:

- Developed endpoints using Golang to handle high-performance connections directly into the core server database.

Tradelink Electronic Commerce Limited - Programmer (6/23 - 9/25)

TDS eDOC Project (OTTO)

Project Description: A specialized document management system built on Tradelink's "Trader Documentation Service" platform for OTTO International (OI). It seamlessly connects an extensive network of supply chain partners (overseas buyers, sellers, and freight forwarders) to facilitate, process, and optimize multi-national document exchange and workflow automation.

Tech Stack: JSP, Java, Eclipse, Tomcat, MySQL, JasperReports, XSL-FO, Cron jobs.

Responsibilities & Achievements:

- Developed and maintained the core modules of the global supply chain system and handled daily production support.
- Used JasperReports and XSL-FO to generate standard PDF and XML business documents.
- Wrote Cron jobs to handle real-time data sync, automated system housekeeping, and log rotations.
- Integrated SFTP workflows to automate document transfers from multinational suppliers and forwarders.
- Built a barcode/QR code generation module and implemented logic to format sequential unit codes (prefix + padded zeros + current number) with rollover handling.
- Added Regex validators to enforce strict data formatting (e.g., specific lengths and counts for HS Code verification).
- Implemented Role-Based Access Control (RBAC) to manage user permissions (create, view, edit, delete, lock) across different hierarchy accounts and teams.
- Created a document template feature allowing users to save completed forms for faster re-use later.

CMHK Kiosk Project

Project Description: A warehouse management system (WMS) kiosk application developed for China Mobile Hong Kong (CMHK), deployed across 4 kiosk units for cargo registration and automated inventory control.

Tech Stack: React Native, Java, Socket.io, MySQL

Responsibilities & Achievements:

- Worked closely with internal teams and frontline staff to design and implement features for daily logistics operations.
- Used Socket.io to build a real-time notification system, connecting client kiosks (Receiving/Shipping options) with warehouse staff units via push notifications and sound alerts.
- Handled asynchronous status transitions to lock the kiosk UI when a staff member is processing an order, preventing duplicate assignments.
- Developed backend verification for scanned QR codes to check order data, formats, and duplicates against the database in real-time.
- Implemented an automated quantity check that cross-references scanned items and highlights the frontend UI (e.g., changing to green) when matching.
- Generated downloadable PDFs via system-created QR codes upon order completion.
- Built a Heartbeat mechanism where each kiosk sends its machine ID via Sockets every 20 seconds, triggering a red alert UI if a unit goes offline.

MY SKILLS

Frontend

- HTML5
- CSS 3
- SCSS
- Tailwind
- TypeScript
- JavaScript
- Vue.js
- React (Next.js)
- Redux
- Recoil
- AJAX
- i18n

Backend

- Node.js
- Express
- Nest.js
- MikroORM
- Knex
- Java
- Golang
- OAuth 2
- MySQL
- MSSQL
- PostgreSQL

DevOps

- Github
- AWS(EC2,S3,Route 53,IAM)
- Docker
- cronjob
- Jenkins

Others

- Git
- RESTful API
- ERD drafting
- Socket IO
- XSL-FO
- JasperReports
- Tomcat
- AI Chatbot Trainer
- Claude Code (agentic AI-assisted development)
- - AI multi-agent orchestration
- - Prompt & workflow engineering (custom skills / hooks)
- - Headless AI task automation (scheduled agents)

EDUCATION

Tecky Academy (12/2022 - 4/2023)

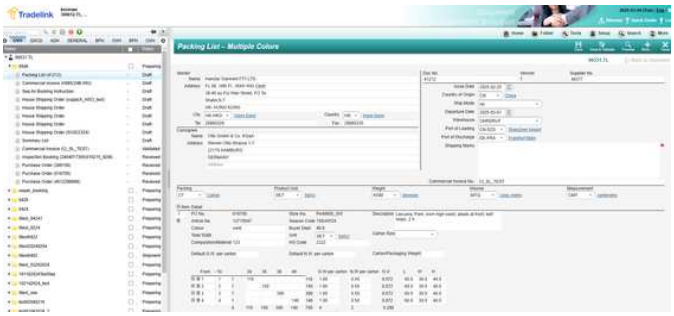
- 16 weeks intensive training on building up programming skill sets and acquired the skill to have my projects finished.

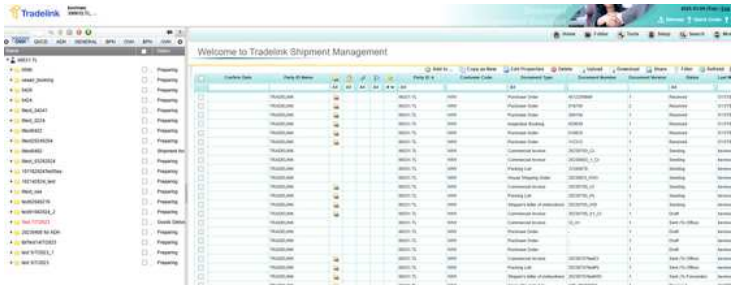
Language

- Cantonese
- English

TDS eDOC Project (OTTO)

- Development, maintenance, and support of the OTTO TDS project, a document management system designed for handling multi-national suppliers and forwarders shipments.
- Ensure system stability and efficiency through continuous improvements and troubleshooting.
- Collaborate with cross-functional teams to enhance system functionalities and optimize workflow automation.
- Provide technical support and solutions to maintain seamless system operations for users.
- https://tds3.tradelinksig.com/scm-sms/home.do

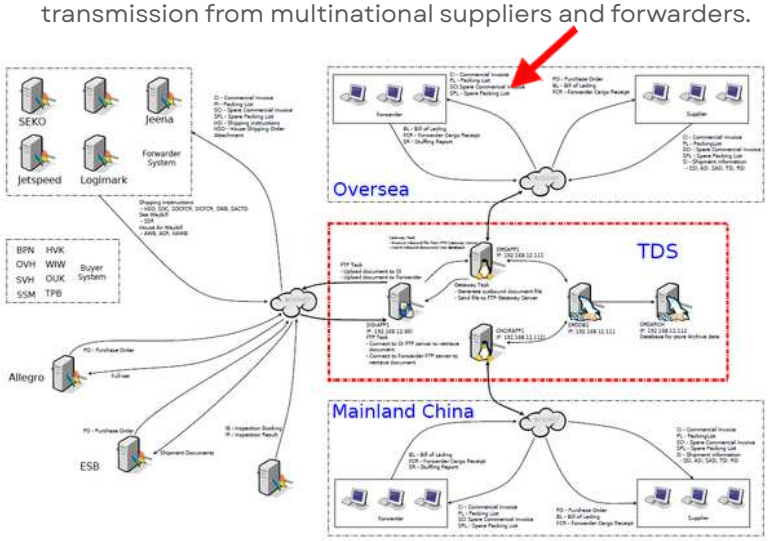




Run cronjob search the dataDesigned and implemented scheduled Cron jobs for real-time data synchronization and system housekeeping. on time.

Integrated SFTP workflows for automated document transmission from multinational suppliers and forwarders.





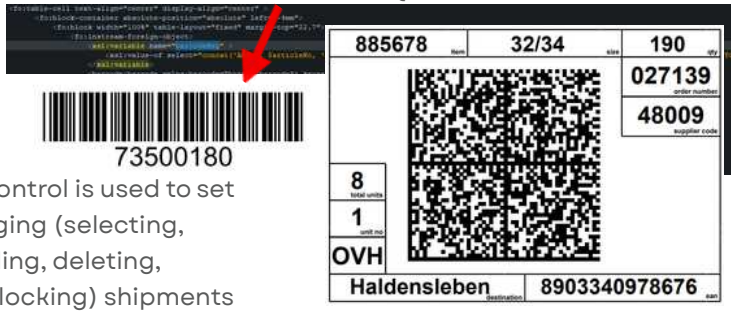
Creating a document based on the items within the shipment for identification.

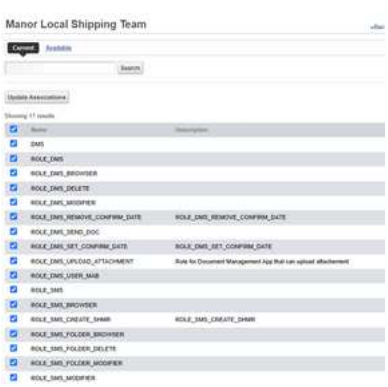


The shipment will be sent before cross-checking those documents.

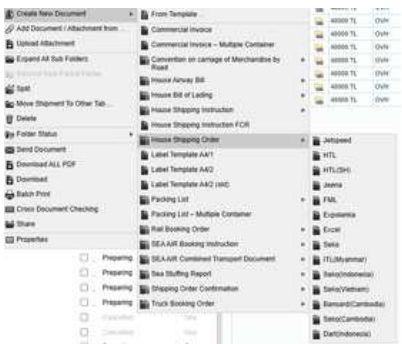


Follow the format to generate the barcode / QRcode.





Group role/user role control is used to set permissions for managing (selecting, creating, adding, sending, deleting, modifying, locking, unlocking) shipments or identifying the shipment flow.
An account can manage many members, a member can add many teams, and a team can add many roles

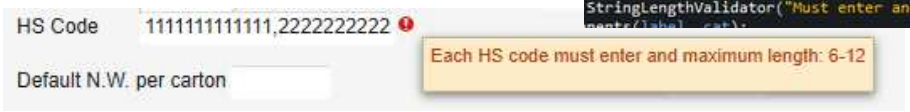


Suppliers can create documents of different versions and different shipping model (sea / air / sea-air / rail / truck).

- Insert the company chop change to base64 then save to SQL.

Set the regular expression to check that the value only accepts the 6-12 length of integer digits and only accepts 5 sets of numbers.

```
"cat_hsCode");
Validator("^([^,]*,){0,4}[^,]*$", "Only accept 5 HS code"));
RegexpValidator("^\\d+(,\\d+)*$","Only accept integer digits"));
RegexpValidator("^\\d{6,12}(,\\d{6,12})*$", "Each HS code must enter and maximum length: 6-12"));
StringLengthValidator("Must enter and maximum length: 6-12", 6, null, false));
```



type	totalDigits	currentNum	startWith	createdAt	lastUpdatedAt
CI BARCODE OVH	8	195	735	[illegible]	2025-05-29 10:30:13
SPL EAN	6	98	5531	2025-05-08 23:35:02	2025-05-29 18:06:19

next barcode will be
735 + 00 + 196

Generate a unit code = (start with) + (fill full digits of zero) + (current number)
the number will be recount if it over the total digits .

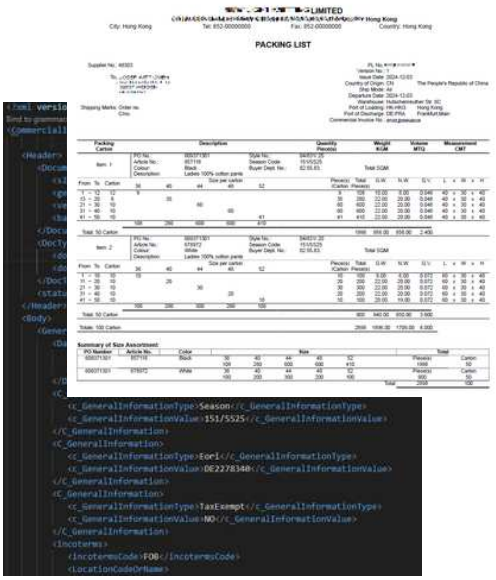
```
@Override //generate the SeqNumber in the document. 10/5/2025
public DataModel genSeqNumber(String type, DataModel dm) {
    int totalEAN = 0;
    final List<LineItem> items = dm.getItems();
    final int size = items.size();
    //count how many item is empty SeqNumber
    for (int i = 0; i < size; i++) {
        final String ean = items.get(i).getEAN();
        totalEAN += (ean == null || ean.length() == 0) ? 1 : 0;
    }

        if(totalEAN<=0) //haven't empty SeqNumber will be return
            return null;
    SeqNum EanNumProfile = this.findByType(type); //find the type of SeqNumber
        if(EanNumProfile==null)
            return null;

        Integer numIncre = EanNumProfile.getCurrentNum();
        Integer totalNumIncre = numIncre + totalEAN;

        //get the length of start with SeqNumber
        Integer startDigits = EanNumProfile.getStartWith() !=null&&!"".equals(EanNumProfile.getStartWith())?EanNumProfile.getStartWith().length():0;
        Integer variableDigits = EanNumProfile.getTotalDigits() - startDigits;
        int overTotalSeqNumber = 0;
        if (totalNumIncre.toString().length() > variableDigits) { //recount the sequence number after total SeqNumber if it is over the total digits
            overTotalSeqNumber++; //overTotalSeqNumber start with 1
            for (Integer k = numIncre; k < totalNumIncre; k++) {
                if (k.toString().length() > variableDigits)
                    overTotalSeqNumber++;
            }
        }
    String StartWith=EanNumProfile.getStartWith()!=null?EanNumProfile.getStartWith():"";
    for(LineItem item:dm.getItems()) { //insert the sequence number to the item
        if(item.getEAN()==null||item.getEAN().isEmpty()) { //find the empty SeqNumber in items
            numIncre++;
            if(numIncre.toString().length() > variableDigits) //recount the SeqNumber when the length of SeqNumber is over then total digits
                numIncre=1;
            //fill full digits of zero => ( start with ) + 0000 + ( current number )
            Integer zeroPatchLength = EanNumProfile.getTotalDigits() - startDigits - numIncre.toString().length();
            String zeroPatch = "";
            if (zeroPatchLength > 0)
                zeroPatch = String.join("", Collections.nCopies(zeroPatchLength, "0"));
            item.setEAN(StartWith + zeroPatch + numIncre.toString());
        }
    }
    //update the SeqNumber into the database, if totalNumIncre is over will take from recounted SeqNumber
    EanNumProfile.setCurrentNum(overTotalSeqNumber == 0 ? totalNumIncre : overTotalSeqNumber);
    EanNumProfile.setLastUpdatedat(new Date());
    seqNumDao.save(EanNumProfile);
    return dm;
}
```

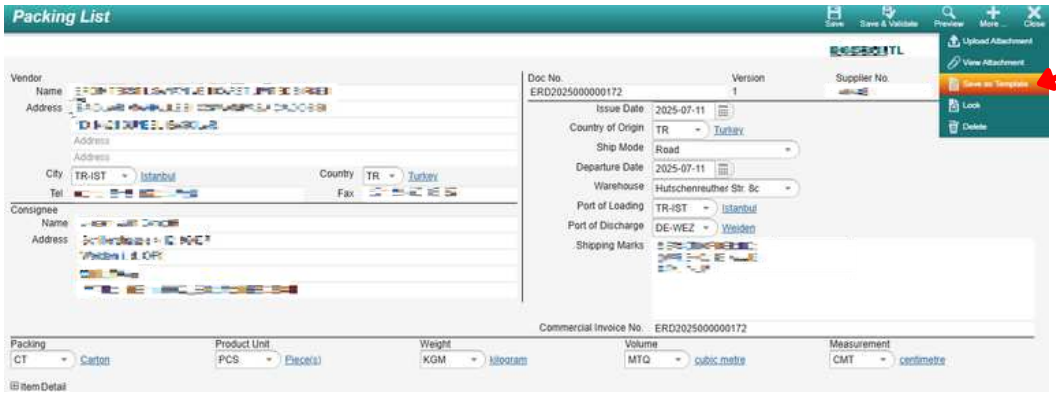
use the JasperReports to generate the PDF and XML



Ability to handle and resolve exceptions in the context of software development.

```
private static boolean hasAllVariable(LineItem i1, LineItem i2, String... fields) {
    for (String field : fields) {
        try {
            //dynamically fetches a method from the LineItem
            java.lang.reflect.Method variable = LineItem.class.getMethod("get" + field.substring(0, 1).toUpperCase() + field.substring(1));
            if (variable.invoke(i1) == null || variable.invoke(i2) == null) return false;
        } catch (Exception e) {
            // if fail to finding the variable in line item return false
            return false;
        }
    }
    return true;
}
```

```
return hasAllVariable(item1, item2, "size", "sealNo", "containerType", "containerNo") && item1.getSize().equals(item2.getSize())
        && item1.getContainerNo().equals(item2.getContainerNo()) && item1.getContainerType().equals(item2.getContainerType())
        && item1.getSealNo().equals(item2.getSealNo());
```

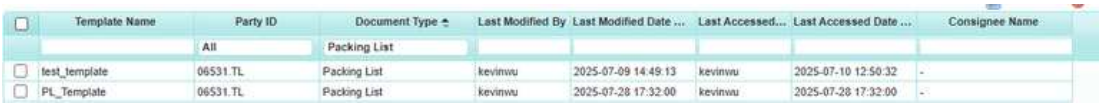


some document can save as template for easily to create another document.

input template name



user can find the template in the template page



Template Name	Party ID	Document Type	Last Modified By	Last Modified Date ...	Last Accessed...	Last Accessed Date ...	Consignee Name
	All	Packing List					
test_template	06531.TL	Packing List	kevinwu	2025-07-09 14:49:13	kevinwu	2025-07-10 12:50:32	-
PL_Template	06531.TL	Packing List	kevinwu	2025-07-28 17:32:00	kevinwu	2025-07-28 17:32:00	-

CMHK Kiosk Project

- Kiosk project is developed with react-native (front end) , socket, Java (back end) and MySQL.
- Responsible for the development, maintenance, and support of the Kiosk project, a warehouse management system that facilitates cargo registration, control, and inventory management.
- Performed system troubleshooting, bug fixing, and performance optimization to ensure stable and reliable operations.

1. The client unit offers three main options: receiving, returning, and shipping.



2. After the client selects an option, for each scanned QR code, the client will call the backend to verify the authenticity of the order (e.g : appropriate format, existence, duplicate scans), then return the order details to the frontend. If all the data is approved, it will be sent to the socket server to notify two staff units.

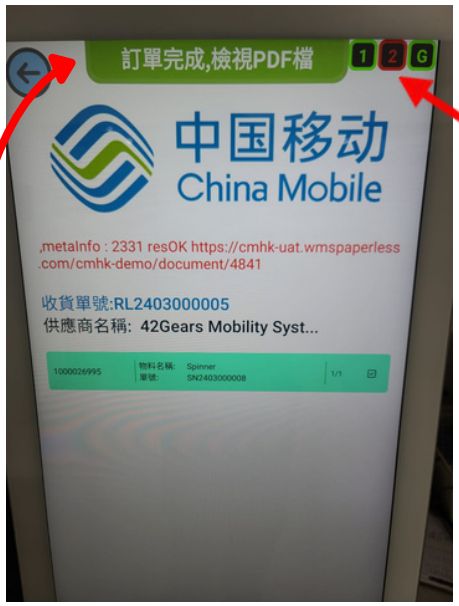
3. An order detail alert and sound notification will simultaneously appear on the two staff units via the socket. Once one unit confirms the order, the other unit will automatically return to its normal display via the socket. Simultaneously, the client unit will receive a response from the staff unit via the socket IO regarding the order. If the states of two staff units are busy or no one is processing the order, the client unit's interface will display a message asking for patience until warehouse personnel confirm. Once the staff unit confirms the order, the response will be relayed to the client unit in the form of a call socket to transition to the order confirmation or rejection page.

4. After confirming the order, the staff units will search for the relevant order from the backend and display it on the frontend.





5. The staff begins to check the sufficiency of each item in the client's checkout order. Within the order interface, it will enter a busy state where no new order notifications from clients will be received until the order is completed. Each scan of a product's code by the staff will automatically verify the quantity until all items in the order are checked. The color green indicates the completion of an item.

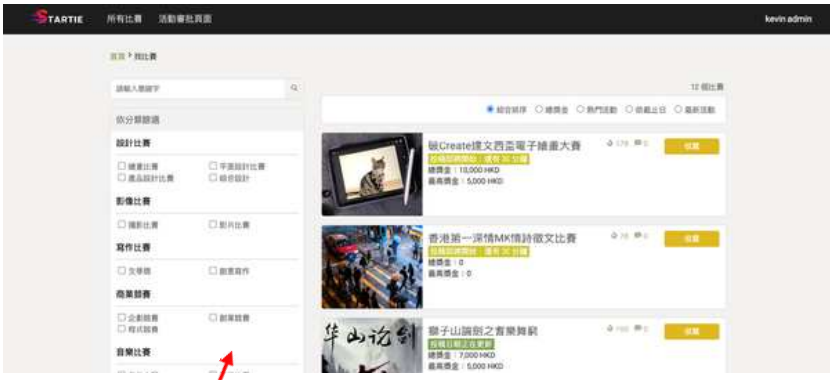


6. After the order is completed, the backend will extract the order number, generate the PDF URL, and send a response to the frontend. This action will prompt the appearance of a PDF view button corresponding to the staff units of the order, enabling staff members to view or download the PDF file through a system-generated QR code.

Each unit is assigned a unique machine ID representing one of three different units, along with a numerical value. Every 20 seconds, the unit increments the numerical value associated with all unit IDs by 1. It then sends its own ID to other units via a socket, resetting its own numerical value to 0 in those units. If the numerical value of a unit is greater than or equal to 3, the staff unit will display the corresponding unit ID in red and show 'connection fail,' achieving the functionality of checking the network status.

Client Project - Startie

Startie is an event display and management platform that equiped with features like create, edit, publish. It also provides different versions of event profile outlook for user to choose.
Startie uses **Stripe** to handle payment. With payment options, events can be set in prior order on display on the event list rendering, and users are enabled to use some of the paid features to customize their events.



The procedures of event setup

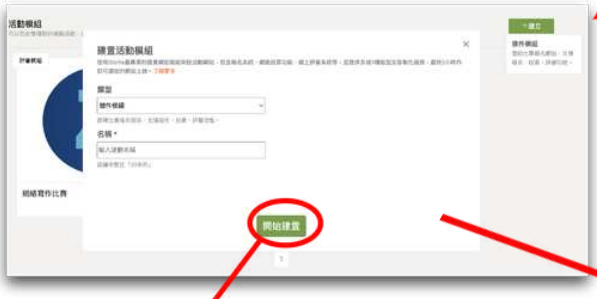


Start with clicking on the account menu on the upper right corner

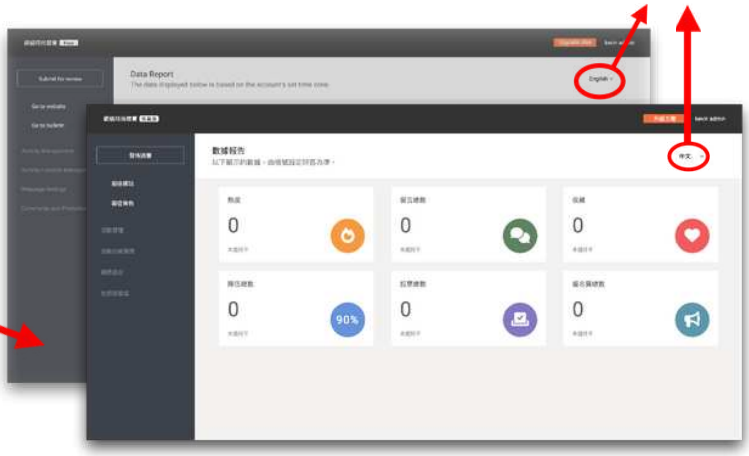
In the event control panel, user can create a “草稿” event.



Events can be filtered by their event types



i18n is implemented for **zh / en** language options





Once the event is created, user can go into the “event dashboard” page to set up all detail and requirement of the event.

Event Customization

Event details such as schedule, activities, descriptions, requirement can be edited with form.
We use **react-hook-form** library for these detail inputs and form submission.

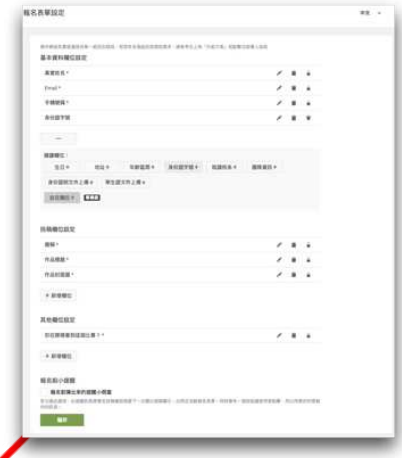
Many CRUD inplementation

As you can see, Startie requires many **crud (RESTful)** logics and operation.
Our backend endpoints also set to be matched with the requirement with the correct **HTTP methods (GET, POST, PUT/PATCH, DELETE)**

In order to prevent issues that caused by time zone differences.
react-timezone-select (library) is used to list out the time zone options.
Therefore it will cause the date inaccurate issue when the backend is hosted in a different time zone.



Beside the basic detail inputs (email, phone number), users can add customized tags to the event.





Endless target group can be generated to accommodate the needs if needed





For event description, library **ckeditor5-react** is used to enable user to customize their outlook of their text inputs.



Endless award can be added and the sum of the awards would automatically calculated.



ckeditor5-react is used here for more description to be displayed on the event profile.



We offer three templates (themes) for user to choose from. User can also update images up to their event profile.



Users can upload the event cover image and add the different tags for easy filter.



Users can edit the FAQ of events.



Users can edit the information of their organization.

All customized plans are being rendered for event joiner to choose.

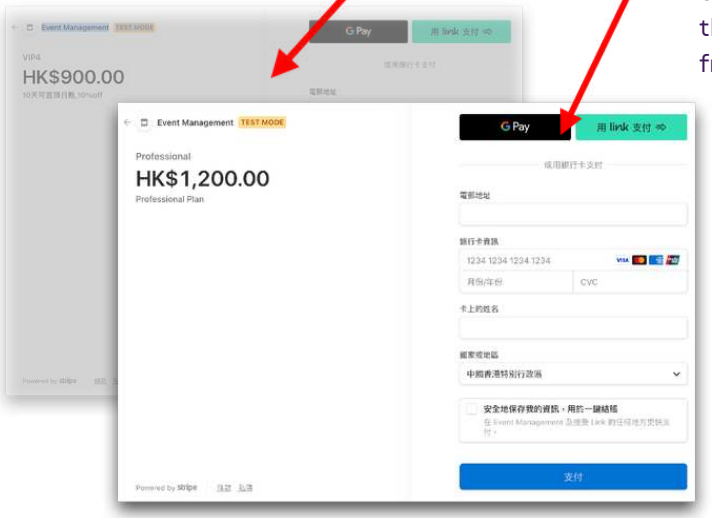






Users can adjust the prices of different VIP plans.

We integrate our our project with the Checkout page from **Stripe**.

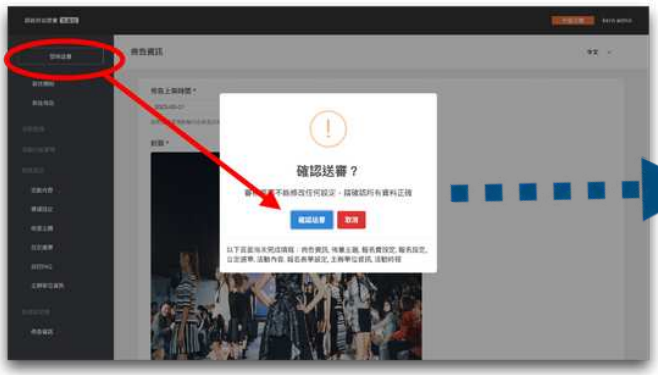






On the other hand, event owner can pay to enable the paid features.
With paid features, users are offered more flexible options to customize their events

After all settings are done, user can submit the event for review.



This project has three types of auth users:

1. **Super admin**
2. **Event Owner**
3. **Event Joiner**



When the event is under review, some of the information can not be modified.



Super admin has a panel where the admin can review, approve and reject the event.

After the event is approved, it will be displayed on the event list where guests can view without logging in.



Click on the event, user will be redirected to the event detail page



Click on this button, user can view the event poster.









Each event has its own data report , visitor number, comment number and submission number can be viewed on the event panel.

Architecture:

